Major Project
(ITM 701)

Project Proposal

Lumen: A Safety-Guarded Conversational AI Companion with Real-Time Crisis Detection

B. E. Information Technology

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Title of the Project: Lumen: A Safety-Guarded Conversational AI Companion with Real-Time Crisis Detection

Domain: AI/ML, Mental Health Technology, Human-Computer Interaction

I. Introduction

Access to timely mental health support remains one of the most persistent gaps in healthcare delivery worldwide. According to World Health Organization data, the treatment gap for serious mental disorders ranges from 35–50% even in developed countries and rises to 76–90% in low- and middle-income countries, with stigma alone linked to an average delay of 9 to 23 years between the onset of a condition and a person's first treatment. Fear of judgment, social devaluation, and the discomfort of naming a problem as "mental illness" consistently rank among the strongest predictors of why people never seek care at all, regardless of country.

In India, this global pattern is compounded by distinctly local barriers. India is home to 18% of the global population and carries a disproportionate share of this burden — mental disorders are the second leading cause of years lived with disability in the country, and nearly 11 out of every 100 Indians live with a mental health disorder, yet between 70% and 92% never receive proper treatment. Unlike a pure resource shortage, a large part of this gap in India is attitudinal rather than infrastructural: lack of awareness about what counts as a mental health concern, fear of being judged or labelled by family and community, and a cultural tendency to treat emotional distress as a private matter to be endured rather than discussed, all lead people to avoid seeking help even where services technically exist. This is precisely why India's own institutional response — Tele MANAS, implemented under the Ministry of Health and Family Welfare with NIMHANS, Bengaluru as its apex nodal body — has been designed as a free, anonymous, 24/7 helpline available in 20 languages rather than a clinic-first model: anonymity and low-friction access are treated as prerequisites to reaching people at all, not conveniences.

As these tools grow more capable and more widely used, the safety infrastructure around them has not kept pace: most existing systems provide either a rigid, clinician-scripted conversation flow with limited natural flexibility, or an open-ended conversational experience with little to no risk-detection scaffolding. Lumen aims to build an end-to-end AI-driven emotional-support platform, relevant to both global and Indian users, that integrates real-time crisis detection, anti-diagnosis guardrails, longitudinal mood-trend analysis, and an explainable safety-decision layer, together with a structured escalation pathway into India's existing helpline and referral ecosystem rather than a generic hotline reference. Using conversational input and journal entries as its data sources, Lumen will

deliver emotionally supportive interaction while continuously screening for risk, enabling safe engagement without overstating its role as a replacement for professional care.

By doing so, the project demonstrates how conversational AI can be engineered for a safety-critical domain, and presents a framework that balances empathetic support, transparency, and responsible boundary-setting between early support and clinical diagnosis — one that is globally applicable in design but explicitly grounded in the language, infrastructure, and cultural context of Indian users.

II. Motivation

Mental health support access is one of the most critical yet under-addressed gaps in personal wellbeing infrastructure, globally and in India alike. High-profile cases — including lawsuits against Character.AI over teen-suicide cases and Italian regulatory action against Replika — highlight the dangers of deploying open-ended conversational AI in emotionally sensitive contexts without adequate safeguards. Existing structured tools such as Woebot and Wysa reduce this risk through scripted flows but sacrifice conversational naturalness and rely on brittle, largely English-first, keyword-based safety triggers. To address these challenges, there is a strong need for an AI system that remains conversationally flexible while continuously screening for risk, tracking patterns over time, and explaining its own safety decisions — in a way that also fits how Indian users actually communicate.

Key Motivations for Lumen:

- **Close a global gap that is sharper in India**: with 70–92% of Indians with mental disorders not receiving treatment, and most private therapy concentrated in metro cities, an AI-first support layer has disproportionate value for tier-2/3 cities and students — a population underserved by existing urban-centric therapy platforms.
- **Enhance Risk Visibility**: apply crisis-pattern detection across both chat and journal content to provide early warnings before a situation escalates.
- **Tackle Gradual Risk Development**: track mood and sentiment trends across sessions rather than relying solely on single-message keyword triggers.
- **Treat Language as a Safety Variable, Not an Afterthought**: Tele-MANAS itself operates in 20 languages precisely because language barriers are documented to cause underutilisation of mental health services in India; Lumen should treat Hindi, Marathi, and code-mixed input as first-class cases for crisis detection and generation, not a stretch goal.

- **Balance Empathy and Safety**: use a constrained-generation approach so conversational responses remain warm and natural while never crossing into diagnostic territory.
- **Mitigate False Positives/Negatives**: layer rule-based detection with an explainability module so flagged (and unflagged) decisions can be reviewed and improved.
- **Anchor Escalation in India's Real Referral Infrastructure**: route flagged users to Tele-MANAS (14416 / 1800-891-4416), campus counselling cells, or other verified Indian services, rather than a subtle in-chat message or a generic international hotline — mirroring how Tele-MANAS itself follows up with high-risk callers instead of treating each contact as a one-off transaction.
- **Respect India's Data Protection Regime**: mental health conversations and journal entries are sensitive personal data under India's Digital Personal Data Protection (DPDP) Act, 2023, which shapes how Lumen handles consent, storage, and retention of chat/journal embeddings from the design stage onward.

III. Problem Statement

Existing AI emotional-support tools — including India-origin platforms like Wysa — force a compromise between safety and flexibility: rigid clinical chatbots stifle conversation, while open-ended companion apps lack safety scaffolding. Safety in both cases is largely restricted to static, English-first keyword triggers evaluated on isolated messages, which is a particularly poor fit for a linguistically diverse user base where distress is often expressed in Hindi, Marathi, or code-mixed phrasing. These systems fail to detect subtle, oblique risk that builds gradually across multiple interactions and siloed journal entries, and when they do escalate, they typically point to a generic or international crisis line rather than India's own Tele-MANAS/District Mental Health Programme referral network. There is a critical need for an architecture that treats safety as a continuous, explainable, multilingual, multi-checkpoint process — one that plugs into India's existing tele-mental-health infrastructure rather than working around it.

IV. Literature Review

Sr. No.	Title of Paper	Review	Analysis / Limitation
1	Delivering Cognitive Behavior Therapy to Young Adults With Symptoms of Depression and Anxiety Using a Fully Automated Conversational Agent (Woebot): A Randomized Controlled Trial	Clinician-designed CBT chatbot that significantly reduced depressive symptoms in a 2-week RCT.	Uses scripted conversations and keyword-based crisis detection, limiting flexibility and contextual safety.
2	An Empathy-Driven, Conversational Artificial Intelligence Agent (Wysa) for Digital Mental Well-Being: Real-World Data Evaluation Mixed-Methods Study	Evaluates Wysa's effectiveness using CBT, DBT, mindfulness, and sentiment analysis. Wysa is itself an Indian-origin platform (Touchkin eServices, Bengaluru), making its gaps directly relevant local precedent.	Journaling is not integrated into safety monitoring, safety-response logic lacks transparency, and escalation is not linked to India's Tele-MANAS/DMHP referral network.
3	Using Psychological Artificial Intelligence (Tess) to Relieve Symptoms of Depression and Anxiety: Randomized Controlled Trial	Shows Tess significantly reduced depression and anxiety symptoms.	Focuses on outcomes rather than safety, with no long-term risk detection or explainability.
4	Systematic Review and Meta-Analysis of Chatbots in the Management of Depressive and Anxiety Symptoms	Meta-analysis of 39 RCTs found chatbots provide modest improvements in depression and anxiety.	Most studies lacked systematic safety monitoring or adverse-event reporting.

V. Gap Analysis

Across reviewed systems, three consistent gaps emerge:

- **Message-Level, Keyword-Based Detection**: Safety detection across all reviewed systems is almost universally limited to individual messages and English-first keyword triggers, with no system described as tracking mood or behavioral trends across multiple sessions, or handling Hindi/Marathi/code-mixed distress language, to inform escalation decisions.
- **Siloed Journaling and Conversation**: Journaling and conversational support are treated as separate, disconnected features — no reviewed competitor routes journal content through the same safety pipeline as live chat, despite both being equally capable of containing crisis-relevant language.
- **Absence of Explainability and India-Anchored Escalation**: Existing tools surface a crisis response when triggered but do not expose to the user or development team which factors contributed to a given risk assessment, and typically escalate to a generic or international hotline rather than India's own Tele-MANAS/District Mental Health Programme referral network, leaving both false positives and false negatives equally opaque.

Lumen directly targets these three gaps through a shared safety pipeline spanning both chat and journal inputs, longitudinal trend-based flagging rather than single-message triggers alone, multilingual (English, Hindi, Marathi) crisis detection, and an explainability layer that surfaces the specific factors behind any detected risk while routing escalation into India's verified referral ecosystem.

VI. Proposed Solution/Methodology

The proposed system, Lumen, aims to build a comprehensive framework that goes beyond scripted chatbot flows and single-message keyword filters. At its core, the solution integrates constrained conversational generation, real-time risk classification, longitudinal trend analysis, and explainability to create a full-stack safety-guarded emotional-support platform, designed for global applicability but grounded in the language and referral infrastructure that Indian users actually have access to. The methodology combines multiple safety techniques while ensuring that flagged decisions remain interpretable and auditable for both users and reviewers.

The first module focuses on input-side risk classification. Using incoming chat messages and journal entries, the system applies rule-based keyword and pattern matching

to screen for crisis-relevant language, chosen deliberately over an opaque ML classifier given the current absence of a labeled clinical dataset suitable for safe, validated training at this stage. This lexicon is extended beyond English to include Hindi and Marathi crisis-language patterns, with allowance for code-mixed (Hinglish) phrasing, reflecting how Indian users typically type rather than assuming English-only input.

To address the issue of context and conversational memory, a retrieval-augmented approach will be adopted. Prior conversation and journal history will be embedded and stored using PostgreSQL with the pgvector extension, allowing the system to retrieve semantically relevant past context rather than relying on raw chat-history truncation, improving both coherence and cost-efficiency of generation.

The next stage of the methodology involves constrained response generation. A language model, scoped by a carefully engineered system prompt, generates supportive responses that are then passed through an anti-diagnosis guardrail — an output-side filter that intercepts and reframes any diagnostic-sounding language before it reaches the user.

To capture risk that builds gradually rather than appearing in a single message, the system includes a mood and trend-tracking module. Sentiment and mood-tag data from journal entries and chat sessions are analyzed over time, flagging sustained negative trends (e.g., N consecutive low-mood entries or declining engagement frequency) for a gentle proactive check-in, distinct from a hard crisis escalation.

Finally, an explainability layer will translate risk-classification outputs into human-readable justification. Instead of simply triggering a generic crisis message, the system will surface the specific factors — keyword matches, sentiment trend, frequency change — that contributed to a given flag, through both an internal audit dashboard and a user-facing escalation interface. This escalation interface routes users to India's Tele-MANAS helpline (14416 / 1800-891-4416) and, where applicable, institutional resources such as a college counselling cell, mirroring Tele-MANAS's own two-tier structure of first-level counsellor support with referral upward rather than inventing a parallel system. All storage of chat and journal data is designed with consent and retention practices aligned to India's Digital Personal Data Protection Act, 2023.

In summary, the methodology combines rule-based multilingual risk classification, retrieval-augmented generation, output-side guardrails, longitudinal trend analysis, and explainability into a unified platform. This holistic approach not only improves detection coverage beyond single-message triggers but also makes the system auditable, transparent, and practically trustworthy for a safety-critical use case, in India and globally.

VII. Hardware & Software Requirements

Hardware:

- **Processor**: AMD Ryzen 7 5700U with Radeon Graphics, 8 Cores / 16 Logical Processors (1.80 GHz base)
- **Memory (RAM)**: 16 GB Installed
- **Graphics Card (GPU)**: Not required for rule-based classifiers; recommended (NVIDIA RTX series or a rented GPU instance) if self-hosting the language model rather than using a hosted inference endpoint
- **Storage**: 256 GB SSD (512 GB preferred for model weights and embedding indices)
- **System**: Windows 10/11 or Linux

Software:

- **Programming Language**: TypeScript, Python (for classifier/NLP components if separated as a service)
- **Frontend Framework**: Next.js 14, Tailwind CSS
- **Voice Interaction**: Web Speech API (SpeechRecognition, SpeechSynthesis)
- **NLP/Classification**: spaCy or NLTK (rule-based keyword/pattern matching), Hugging Face Transformers (sentiment/emotion scoring)
- **Embeddings**: sentence-transformers (for pgvector-backed retrieval)
- **Language Model**: Open-weight LLM served via Ollama or Hugging Face Inference Endpoints
- **Database**: PostgreSQL with the pgvector extension; Redis for session state/job queue
- **Testing & Tooling**: Jest (guardrail/classifier logic), Git/GitHub, VS Code

VIII. References

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